Let the Thermal Camera Decide When to Adapt: Infrared-Gated Test-Time Adaptation for Visible–Infrared Maritime Detection

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***Abstract*—Visible cameras on surface vessels degrade in fog, low light, glare and noise, and so do detectors trained on clean frames. Re-estimating its batch-normalization statistics at test time needs no label, but adapting undegraded frames costs up to 0.031 average precision (AP) at night. The visible camera cannot decide where to adapt: a novelty score of its own image statistics flags every clean night frame as abnormal. We let a co-mounted thermal camera decide. IR-gated batch-normalization adaptation (IG-BN) re-estimates the visible detector's statistics on eight-frame blocks only where the visible frame looks abnormal and a normal-looking thermal frame reads daylight. It takes no gradient, changes no weight and has a causal form. On the Pohang Canal dataset, a canal and port route, every variant was pre-registered before it was scored. IG-BN passed a 60-condition development grid with no condition worse. On one held-out daylight run it replicated its development gains in ship AP50-95 under simulated fog (+0.037), low light (+0.019) and, with a glare term, glare (+0.031); under sensor noise, absent from development, it gained +0.182 in a first test. Inside a ten-pass ensemble the gains add under noise and fog. Each held-out condition corrupts one camera only, so the held-out run tests the visible half of the gate; the day-night decision rests on development night, and whether a clock could replace the thermal camera there is untested. Earlier uses of the second sensor, as an uncertainty-weighted voter and as a veto, did not survive registration.**

***Index Terms*—Batch normalization, infrared imaging, maritime object detection, sensor fusion, test-time adaptation, thermal imaging, unmanned surface vehicles, visible–infrared detection**

I. Introduction

Maritime perception for autonomous and crewed surface vessels increasingly pairs a visible camera with a long-wave infrared (thermal) camera, on the assumption that the two fail in different conditions. A missed vessel or buoy is a collision risk, and the conditions that hide one, fog, darkness, glare and sensor noise, are the conditions in which a collision-avoidance system most needs its cameras. Visible imagery degrades in fog, haze, glare, rain, darkness and sensor noise. Thermal imagery sees through darkness and glare but fails at thermal crossover, when vessel and water reach the same temperature. Neither failure announces itself. A visible detector trained on clean frames meets a fogged or underexposed frame with activation statistics its BatchNorm layers were not fitted to, and simply returns fewer and worse boxes.

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Appendices A–K, cited in the text, are provided as supplementary material.

Test-time adaptation of normalisation statistics is the cheapest repair available. Replacing or mixing the BatchNorm statistics with those of a test batch [1, 2] needs no label, no gradient and no change to any weight, and it recovers a large part of the loss under common corruptions. Its difficulty is not how to adapt but where. Adapting a frame that was never shifted moves a well-fitted detector away from its training statistics. In an unregistered screen, BN-mix applied to every frame gained on corrupted day frames but cost 0.007–0.008 ship AP on clean day frames and up to 0.031 on night frames (Section V-A). Selective test-time adaptation methods decide where to adapt from the model's own outputs or features: entropy, the divergence of one modality's prediction from the fused one, or feature statistics against a clean reference (Section II). A single visible camera faces an ambiguity here that none of these signals resolves. A dark frame may show a dark scene, which the detector was trained on, or a degraded camera, which it was not. The visible detector's own health score cannot tell them apart: on our development set it flags all 1,032 clean night frames as unhealthy and none of the 1,200 clean day frames (Section IV-A).

A thermal camera answers the question with a different sensor. Our method, IR-gated BatchNorm adaptation (IG-BN), re-estimates the visible detector's BatchNorm statistics only where three things hold: the visible frame reports itself unhealthy, the infrared frame reads daylight, and the infrared frame is itself healthy (Fig. 1). A dark or veiled visible frame that the thermal camera places in daylight is a degraded camera, not night, and it is adapted. A dark frame the thermal camera places at night is left alone. The thermal reading we use is a statistic of the thermal image after the dataset's per-frame 8-bit stretch, not a radiometric measurement, and a clock or the visible camera's exposure metadata could carry part of the same day–night information; we discuss both in Section VII.

We arrived at this design by elimination, on the same two-stream system. The system was built so that predicted box uncertainty would weight the fusion of the two streams. A pre-registered test found that uncertainty informative about which boxes are right but not useful for fusing them (Section VI-A). An image-statistic veto that dropped the visible stream at night shipped instead, and it cost 0.1847 ship AP at night once the visible detector was retrained on corrected labels (Section VI-B). No learned gate we registered beat switching that veto off (Section VI-C). Through all of it, the thermal stream added little to the boxes when scored against the

visible stream's labels, the only labels the held-out run has (Section VII).

Our contributions are:

1) **IR-gated test-time adaptation for visible–infrared detection.** A label- and gradient-free adaptation of the visible detector, gated per frame by the thermal camera's reading of illumination and by both streams' frame health (Section IV). It has a causal form for live streams (IG-BN-causal), a glare term (IG-BN+glare), and composes with a ten-pass ensemble (IG-BN+ens). We found no prior test-time adaptation for visible–infrared detection in which the second sensor decides, without gradients, whether to adapt the first (Section II).
2) **Pre-registered evidence on development and one held-out run.** Every variant was registered before any of its numbers existed. Six passed development on 60 cells with no cell worse, one (IG-BN-stale) failed and was dropped, and all six were then scored on a held-out daylight run, where none was worse than its baseline on any cell. Under fog (+0.0370 ship AP), low light (+0.0186) and glare with the glare term (+0.0309), the held-out gains replicate development. Under sensor noise (+0.1823 [+0.1733, +0.1916]), rain and blur, conditions the development grid did not contain, they are first tests (Section V).
3) **Negative results that locate where the second sensor helps.** On the same system, predicted uncertainty is informative but not useful for fusion; a sensor-selection veto encodes a claim about the detector and breaks when the detector changes; and frame-level gates pay a score-pooling penalty that a fixed ordering avoids (Section VI).

Adverse conditions are simulated, from a camera-chain model for fog, low light and glare and from ImageNet-C-style severities for noise, rain and blur. The data are from a canal and port route, not open sea. The held-out run is all daylight, and each of its cells corrupts one stream at a time, so it tests the gate's visible term, not its day–night decision (Section VII).

II. Related work

Test-time adaptation through normalisation. AdaBN [1] replaces BatchNorm statistics with those of the test batch. BN-mix [2] mixes test statistics with the source statistics at a prior strength N; we use it unchanged. DUA [3] updates the normalisation statistics online from a few test samples with a decaying momentum, and is evaluated on detection as well as classification. TENT [4] trains the BatchNorm affine parameters by minimising entropy, and ActMAD [5] aligns activation statistics at many layers by gradient and applies to detectors. Adapting where nothing shifted can hurt, and per-sample selection of what to adapt is established: SuMi [6] adapts only samples that pass feature-quartile and entropy tests, and updates normalisation affine parameters by gradient.

Multimodal test-time adaptation. Several methods adapt only the shifted modality, or reweight toward the reliable one. Each decides from the model's own predictions or features, and each updates parameters by gradient. READ [7] modulates cross-modal attention with a confidence-aware entropy loss. TSA [8] learns modality-specific adapters and a router that chooses which modality to adapt, and reports negative transfer when an unshifted modality is adapted, the failure our gate guards against. AdaPGC [9] flags the corrupted modality per sample by the divergence of its prediction from the fused prediction. CARAT [10] flags each sensor stream against its own clean statistics, and argues that cross-modal agreement misleads when sensors measure different physical processes. DASP [11] flags the biased modality by the redundancy of its fused features. ABPEM [12] bootstraps cross-attention from self-attention under a principal-entropy loss, and MM-TTA [13] adapts camera–LiDAR segmentation with cross-modal pseudo-labels. These are evaluated on audio–video classification, sentiment, image–text classification, wearable time series and segmentation. Night-TTA [14] is the closest in sensors: it adapts night colour–thermal segmentation by training BatchNorm affine parameters against an ensemble teacher, on every night frame, without deciding which frames to adapt.

Visible–infrared detection with illumination or uncertainty. Illumination-aware fusion is established in multispectral pedestrian detection. IAF R-CNN [15] weights its colour and thermal sub-networks by an illumination estimate from the colour image, and MBNet [16] uses illumination to reweight modality features and aligns them. ProbEn [17] fuses colour and thermal detections by probabilistic ensembling. UA-CMDet [18] weights each object's training loss by a rule built from cross-modal box overlap and RGB illumination, and at inference multiplies the RGB branch's scores by an illumination weight before non-maximum suppression. Zhao et al. [19] weight the visible and infrared branch losses by a label uncertainty estimated during training. All of these use illumination or uncertainty to set fusion weights, not to adapt a detector. Gaussian YOLOv3 [20] attaches per-coordinate variance to a single-stage detector; we use its head (Section VI-A).

Maritime detection. Maritime YOLO variants such as RDSC-YOLOv4 [21] and YOLOv7-Sea [22] report accuracy without uncertainty or adaptation. Benchmarks for unmanned surface vehicles, MaSTr1325 [23], MODS [24] and LaRS [25], are visible-only. The Pohang Canal dataset [26] with PoLaRIS annotations [27] provides paired visible–thermal maritime video with boxes on both streams; the Singapore Maritime Dataset [28] pairs visible with near-infrared video that need not show the same scene, and MassMIND [29] is thermal only.

What differs here. We found no test-time adaptation method for visible–infrared object detection among the papers above. Ours differs from the multimodal methods in three ways. The thermal camera decides *whether* to adapt the visible detector, and it decides from a statistic of its own image, not from agreement between model outputs. No gradient is taken and no weight changes. And every variant was registered before it was scored. Selective adaptation itself, adapting only the corrupted modality, is not our contribution, and neither is illumination-aware gating as such, which IAF R-CNN and UA-CMDet apply to fusion weights.

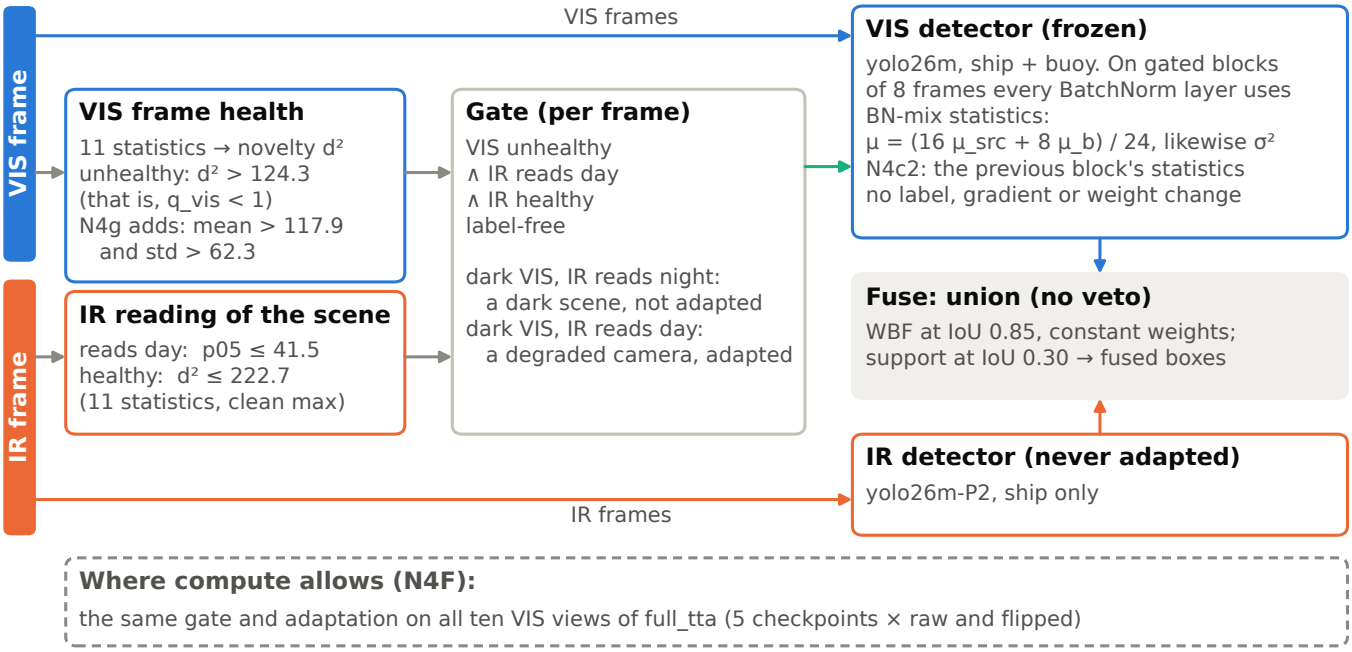


Fig. 1. **IR-gated BatchNorm adaptation (IG-BN).** Per frame and without labels, statistics of both cameras decide whether the frozen visible detector normalises with BN-mix statistics re-estimated on a block of eight frames; the thermal detector is never adapted and the streams are fused by `union`.

III. Setting

A. Data

The Pohang Canal dataset [26] records a 7.5 km route through a canal, a port and near-coastal water, not open sea, with a visible camera at 2048×1080 and a thermal camera at 640×512, both at 10 Hz. PoLaRIS [27] supplies ship and buoy boxes on both streams. There are five runs: pohang00 (day), pohang01 (night), and pohang02 to pohang04 (day). On disk they hold 158,319 images, 1,183,736 boxes and 28,388 paired visible–infrared frames. The two cameras are not co-registered: frames are paired by nearest timestamp, and the residual misalignment, measured, is 3–6 px median per run. Visible frames are letterboxed to 640×640; thermal frames are delivered as 8-bit images by per-frame min–max normalisation, and the 14-bit radiometric values are not recoverable. Splits are interleaved contiguous blocks with guard bands, identical across modalities, and the night labels are the corrected set of a registered restore (Appendix A).

B. Detectors and fusion

Both streams use YOLO26-m [30]: the visible detector with two classes, the thermal detector with a single ship class and a P2 neck, because it cannot see buoys. The backbone was chosen on a 93-run benchmark of 31 YOLO variants from six families with three seeds each (Table S15, Fig. S7): the YOLO26 m, l and x variants form a top tier whose differences are below their seed standard deviations, and yolo26m is the fastest of the three. We use five independently trained system pairs (visible seed k with thermal seed k), trained on the corrected labels with the held-out run removed from every list, and report the mean over the five.

The baseline fusion is `union`. Both streams pass to Weighted Boxes Fusion [31] at IoU 0.85 with constant stream weights (the visible weight, 0.9926, is what the designed uncertainty weighting measured to on every frame, Section VI-A), and a box whose other-stream partner overlaps it at IoU 0.30 has its score raised by half. At this threshold only 0.05 percent of visible boxes find a thermal partner, so the fusion is concatenation with a support bonus. The shipped system, `shipped`, is `union` plus a veto that drops the visible stream when the thermal camera votes night and the visible frame reads dark or veiled (Appendix D.2). A registered study of replacements for that veto found that switching it off improves on it and that no gate we registered beats `union` (Section VI-C); `union` is therefore the baseline every adaptation is measured against. `full_tta` is the strongest system that changes no detector: five checkpoints per stream, each run on the raw and the horizontally flipped frame, merged per stream by WBF at IoU 0.55, with non-local-means denoising of thermal frames that fail their health check, and no veto. It costs ten detector passes per stream per frame.

C. Corruptions

Adverse conditions are simulated on the content rows of each frame. Fog, low light and glare follow a camera-chain model in physical order: atmosphere, optics, sensor noise, exposure, the dataset's own infrared stretch and quantisation. Fog is Koschmieder attenuation in linear radiance on per-frame metric depth [32] from Depth Anything V2 [33]. Low light is exposure scaling to the real night level, with the night noise measured on training frames. Glare is a source placed against the frame's computed horizon, with spread and exposure response fitted to the one real sun-entry event in

training. Sensor noise, rain and motion blur are ImageNet-C-style severities [34] with no physical anchor. An earlier off-the-shelf corruption set was found broken at the source (its fog was a fixed 21-px blur) and retired (Appendix B). Sun glint, spray and salt on the lens were not modelled.

D. Evaluation and statistics

Terms. A *cell* is one combination of visible condition, thermal condition and, on development, day or night. A *draw* is one random realisation of a corruption. *The claim* is the fusion property fused $\geq$ max(VIS, IR): the fused output scores at least as well as the better single stream.

Development. 2,232 paired validation frames, 1,200 by day (pohang00, 02, 03) and 1,032 by night (pohang01). The grid crosses five visible conditions (clean, fog at severities 1 and 2, low light, glare) with six thermal conditions (clean, fog at severities 1–3, noise at severities 2–3), by day and by night: 60 cells. It contains no visible noise, rain or blur. Corrupted cells average three draws per system; a cell's value is the mean over the five systems, with a 95 percent t-interval (df 4) on paired deltas.

Held-out. pohang04, an all-daylight run of 12,482 frame pairs, was never scored by any fusion system before a single pre-registered look, and none of the five system pairs or their reference statistics saw it. It has no thermal labels, so every held-out number is scored against the visible ground truth (156,652 boxes). Nine cells corrupt one stream at a time with four fresh draws each (visible fog, low light, sensor noise, rain and blur with clean thermal; clean visible with thermal fog, noise and blur; and clean/clean), and a tenth, visible glare, was added for the glare term. Held-out intervals are paired moving-block bootstraps over frames (block 20 frames, 1,000 replicates) around the five-system mean, because frames at 10 Hz are autocorrelated and a frame-level bootstrap halves the true width (Appendix C.3). They carry frame-sampling variance only, not variance over systems or draws, so they are not directly comparable with the development t-intervals.

Metric and rule. The metric is ship AP50-95 (local linear-interpolation AP), the only class both detectors emit. One rule decides every registered verdict, on development and held-out data alike: an arm passes (development) or replicates (held-out) if no cell is worse than its baseline by 0.0014 or more with the interval below zero, and at least one cell is better by 0.0060 or more with the interval above zero. The 0.0014 floor is the measured two-sigma noise of a paired delta; 0.0060 is the floor widened by the measured autocorrelation inflation (Appendix C.2–C.3). On held-out cells the development grid did not contain, the same rule is a first test, not a replication.

Registration. Each arm's prediction, cells, statistic and rule were written and committed before any of its numbers existed. No constant, threshold or calibration of any system was fitted on held-out labels, but two gate constants were chosen with the held-out run's label-free statistics (Section IV-D), and no untouched test set existed before it (Appendix C.1). It is scored only through registered tests, and each scoring is logged in an exposure ledger. It has been scored nine times: the original look of the shipped system, a re-score of its corrupted cells after the corruption model was replaced, and seven later registrations, six of them the adaptation tests of Section V-B (Appendix G). The evidence record names the variants by registration codes (IG-BN = N4, IG-BN-causal = N4c2, IG-BN-stale = N4c, IG-BN+glare = N4g, IG-BN+ens = N4F, IG-BN+ens+glare = N4Fg, IG-BN+AE = N4ae).

IV. Method

A. The ambiguity a single camera cannot resolve

A degraded visible frame raises a question its own pixels cannot answer: is the scene dark, or the camera? The visible health score `q_vis` is the clean-frame bound (124.3) divided by the Mahalanobis novelty d^2 of eleven frame statistics (gradient Gini, the ratio of Laplacian to variance, spectral slope, texture cover and seven others), fitted on clean day frames of the development runs and capped at 1, so `q_vis` < 1 means $d^2 > 124.3$. It is a good detector of an unusual visible frame: it flags fogged, underexposed and noisy day frames. It is also, necessarily, a detector of night. On the development paired set it flags all 1,032 clean night frames as unhealthy and none of the 1,200 clean day frames. Night frames are in the visible detector's training distribution, and adapting them hurts (Section V-A). The thermal image's 5th-percentile intensity separates the two cases on development: clean day frames lie between 2 and 34 and clean night frames between 49 and 113 on its 8-bit scale. Because each thermal frame is stretched to its own minimum and maximum, this is a statistic of the stretched image, not of scene temperature; we have not established how much of the separation comes from the stretch.

B. The gate

Per frame, without labels, the gate fires when three conditions hold:

- the visible frame reports itself unhealthy: `q_vis` < 1;
- the thermal frame reads daylight: its 5th-percentile intensity `ir_p05` $\leq$ 41.5, the midpoint of the empty margin between clean day and clean night on development;
- the thermal frame is itself healthy: the Mahalanobis novelty of its own eleven frame statistics is within 222.7, the maximum over clean development frames (the *merge bound*).

The third term exists because a degraded thermal stream can misread the scene: after the per-frame stretch, thermal fog and noise move a frame's 5th percentile across the day–night threshold (Appendix D.5), and without the health term the gate leaks onto up to 2.2 percent of night frames under thermal fog. With it, the gate fires on no night frame under any thermal condition on development. The threshold 41.5 was fitted on the same development frames on which this separation is shown.

C. The adaptation

Frames are taken in non-overlapping blocks of eight consecutive frames. Where the gate fires, every BatchNorm layer of the visible detector normalises with the BN-mix statistics of Schneider et al. [2]:

$\mu = (N\mu_{\mathrm{src}} + n\mu_b)/(N+n)$, $\sigma^2 = (N\sigma^2_{\mathrm{src}} + n\sigma^2_b)/(N+n)$,

with N = 16, n = 8, and μ_b, σ_b^2 the block's statistics over its frames and spatial positions, computed layer by layer as the block passes through the network. Frames the gate does not pass keep the detector's plain output. No gradient is taken, no weight changes, no label is read, and the thermal detector is never adapted; the two streams are then fused by `union`.

D. Variants

- **IG-BN-causal.** IG-BN normalises a block with its own statistics, which include up to seven later frames (0.7 s at 10 Hz). IG-BN-causal normalises each block with the statistics IG-BN measured on the previous block, at the cost of a second pass per block. For a live stream it is the version to use. A first causal variant, IG-BN-stale, reused the statistics measured during the previous block's own adapted pass; it failed by construction (Section V-A).
- **IG-BN+glare.** IG-BN's gate almost never fires under glare, because a glared frame is bright and textured and does not look novel to `q_vis`. IG-BN+glare also fires, under the same two thermal terms, when the visible frame's mean intensity exceeds 117.92 and its standard deviation exceeds 62.31, each the maximum over clean development day frames.
- **IG-BN+ens.** The same gate and adaptation applied to every visible view of `full_tta` (ten passes per stream). IG-BN+ens+glare uses IG-BN+glare's gate.
- **IG-BN+AE.** Gated frames that are dark (mean linear luminance below 0.18, auto-exposure's own target) take the detections of an auto-exposed frame instead.

Where the constants come from. N = 16 and the gate's terms were chosen on a screen draw before any development verdict; N = 64 halved both the gains and the losses. The visible health bound, the thermal day threshold and the thermal health bound are the frozen constants of the shipped system, fitted on clean development frames. Two choices used the held-out run's label-free frame statistics before any of its scores existed. The thermal health term was switched from the shipped system's tighter *authority bound* (the 99th percentile of the same novelty over clean development frames) to the merge bound after the former was seen to pass only 9.5 percent of clean held-out thermal frames (the merge bound passes 86.7 percent). The glare thresholds were checked to fire on 0.10 percent of clean held-out frames, after brightness percentiles that separate development glare perfectly were seen to fire on 55 percent of them. No constant was fitted on held-out labels.

V. Results

A. Development

Ungated adaptation hurts. In a screen on one corruption draw (two systems, not registered), BN-mix with N = 16 on every frame gained +0.0635 and +0.0767 ship AP on low-light day and +0.0088 and +0.0083 on fog by day, and cost 0.0080 and 0.0068 on clean day frames and up to 0.0311 at night. The gate is designed to keep the first and avoid the second. No registered test has compared the gate with ungated adaptation or with a gate on one sensor alone, so the losses the gate avoids rest on this screen.

IG-BN passes development (Table I). No cell of the 60 is worse than `union`; the worst Δ is -0.00001. Five are better beyond 0.0060 with intervals above zero: low light by day +0.0706 [+0.0593, +0.0818], low light with thermal fog at severity 1 +0.0357, fog at severity 1 +0.0094, fog +0.0091 and low light with thermal fog at severity 2 +0.0069. The gate fires on 16 cells, all by day: on 100 percent of low-light day frames and 99.6 percent of fogged day frames with clean thermal, and on no night frame under any thermal condition, so IG-BN equals `union` on every night cell. As thermal fog thickens the gate fires less, because the thermal stream stops vouching for daylight: on low-light day frames it fires on 42.5, 14.9 and 4.5 percent at thermal fog severities 1–3, and on none under thermal noise. Where fog degrades both bands, as real fog does, the gate therefore adapts few frames.

The variants. IG-BN+glare gains +0.0240 under glare by day and +0.0130 with the thermal stream in light fog; its worst cell is -0.0008. IG-BN-stale collapses from the first block that uses stale statistics, because they were measured under that block's own stale normalisation rather than IG-BN's: mean maximum confidence falls to 0.21 against IG-BN's 0.75, and it was not taken further. IG-BN-causal passes; against IG-BN the price of causality is -0.0038 [-0.0045, -0.0031] on low-light day and within ± 0.0003 on fog. IG-BN+AE gains +0.0165 [+0.0053, +0.0278] over IG-BN on low light by day; darkness splits the conditions exactly, since every gated low-light frame is dark and no gated fog frame is, so on fog IG-BN+AE is IG-BN. Inside the ensemble, IG-BN+ens's gain survives almost undiminished (+0.0696 on low-light day, on its own draws), and so does the glare gain (IG-BN+ens+glare, +0.0179 on glare by day). IG-BN+ens+glare's rule had no gain requirement: no cell worse than `full_tta`, and none below IG-BN+ens by 0.0014 where the glare term adds frames. The held-out build runs in half precision with fused batch statistics; a development check priced that at no more than 0.0003.

B. Held-out run

Each variant that passed development was registered for the held-out run before any of its held-out numbers existed. None is worse than its baseline on any held-out cell (Table II). Fog, low light and glare repeat development conditions and are replications; sensor noise, rain and blur are first tests.

IG-BN. No cell is worse than `union`, and three are better beyond 0.0060 with intervals above zero. Under visible sensor noise, re-estimating the normalisation statistics takes the fused output from 0.0169 to 0.1992; the ten-pass `full_tta` reaches only 0.0206 there. This is the established benefit of BN re-estimation under noise [2], and a first test, since development has no visible noise. Fog gains +0.0370 and low light +0.0186, as on development. On the four cells with a clean visible stream (thermal clean, blurred, noisy or fogged) the gate fires on no frame, so IG-BN is `union` there. Because every held-out frame is daylight and every adaptation cell has a clean thermal stream, these cells exercise the gate's visible term; its day–night term is tested only on development night (Section V-C).

IG-BN-causal. The causal version keeps IG-BN's held-out gains, with no cell worse: against IG-BN it costs at most

TABLE I
REGISTERED VARIANTS ON DEVELOPMENT: 60 CELLS, FIVE SYSTEMS, SHIP AP, PAIRED AGAINST THE BASELINE NAMED

Variant	Change	Baseline	Better	Worse	Largest gain [95% CI]	Verdict
IG-BN	gate + BN-mix	`union`	5	0	low light, day: +0.0706 [+0.0593, +0.0818]	PASS
IG-BN+glare	+ glare term	`union`	2	0	glare, day: +0.0240 [+0.0178, +0.0302]	PASS
IG-BN-stale	stale statistics	`union`	0	10	low light, day: −0.1637	FAIL
IG-BN-causal	previous block	`union`	4	0	low light, day: +0.0668 [+0.0559, +0.0777]	PASS
IG-BN+AE	+ auto-exposure	`union`	5	0	low light, day: +0.0871 [+0.0734, +0.1008]	PASS
IG-BN+ens	in `full_tta`	`full_tta`	5	0	low light, day: +0.0696 [+0.0620, +0.0757]	PASS
IG-BN+ens+glare	+ glare term	`full_tta`	2	0	glare, day: +0.0179 [+0.0141, +0.0207]	PASS

Better and worse count cells beyond the rule's floors (Section III-D); IG-BN+ens and IG-BN+ens+glare were scored on a second set of draws, with moving-block intervals because their arms are five-system ensembles.

TABLE II
THE HELD-OUT RUN (POHANG04), ONE CORRUPTED STREAM PER CELL, THERMAL STREAM CLEAN IN EVERY CORRUPTED ROW: SHIP AP AGAINST THE VISIBLE GROUND TRUTH

VIS condition	Gate	`union`	IG-BN − `union`	IG-BN-causal − `union`	`full_tta`	IG-BN+ens − `full_tta`
noise s2 (first test)	85.2%	0.0169	**+0.1823** [+0.1733, +0.1916]	+0.1820 [+0.1730, +0.1913]	0.0206	**+0.2174** [+0.2079, +0.2265]
fog	67.8%	0.0445	**+0.0370** [+0.0328, +0.0408]	+0.0363 [+0.0323, +0.0401]	0.0586	**+0.0445** [+0.0401, +0.0488]
glare s2 (+glare)	81.0%	0.1417	**+0.0309** [+0.0290, +0.0327]	—	0.1656	**+0.0311** [+0.0286, +0.0332]
low light	85.2%	0.2152	**+0.0186** [+0.0154, +0.0219]	+0.0180 [+0.0148, +0.0212]	0.2458	**+0.0190** [+0.0152, +0.0232]
rain s2 (first test)	18.6%	0.2209	+0.0036 [+0.0025, +0.0048]	+0.0035 [+0.0023, +0.0047]	0.2534	+0.0033 [+0.0022, +0.0047]
blur s3 (first test)	9.2%	0.0419	+0.0001 [+0.0000, +0.0001]	+0.0001 [+0.0000, +0.0001]	0.0530	+0.0000 [+0.0000, +0.0001]
clean (4 cells)	0.0%	0.2635–0.2682	0	0	0.2923–0.2936	0

Mean over the five systems and the four draws; Δ is paired against the arm's own baseline with a moving-block 95 percent interval, and Gate is the share of frames the gate passes (IG-BN+glare's gate on the glare row).

0.0007 (fog) on the held-out run and 0.0038 on development low light.

IG-BN+glare. None of the nine registered cells corrupts the visible stream with glare, so its registration added a tenth. There IG-BN+glare gains +0.0309 [+0.0290, +0.0327] over `union` (0.1417 to 0.1726), where IG-BN alone gains nothing, as on development; in the three other cells where the glare term adds two or three gated frames per draw, IG-BN+glare equals IG-BN to the fourth decimal.

IG-BN+ens. Inside the ensemble, no cell is worse than `full_tta`. The gain is larger than alone under noise (+0.2174 against +0.1823; 0.0206 to 0.2380) and fog (+0.0445 against +0.0370; 0.0586 to 0.1031), and the same under low light (+0.0190 against +0.0186) and rain (+0.0033 against +0.0036). Under glare, IG-BN's gate fires on 1.9 percent of frames and IG-BN+ens equals `full_tta`, so glare was registered separately with the glare term (IG-BN+ens+glare, after a development pass). It replicates: +0.0311 [+0.0286, +0.0332] over `full_tta` (0.1656 to 0.1967), +0.0240 above one-pass IG-BN+glare, with no other cell worse than IG-BN+ens. On five of the six held-out cells with a corrupted visible stream, IG-BN+ens (IG-BN+ens+glare on glare) is the highest-scoring system we measured; under blur it ties `full_tta`.

IG-BN+AE, with a cost development could not show. Low light gains +0.0259 [+0.0231, +0.0285] over `union`, +0.0073 [+0.0046, +0.0099] over IG-BN, as on development. But the held-out rain darkens every gated rain frame below auto-exposure's threshold, so auto-exposure replaces adaptation there and IG-BN+AE on rain is 0.0036 below IG-BN. Development has no rain. We do not recommend IG-BN+AE for that reason.

C. Where the gate does not fire

The gate's job is as much where it stays off as where it fires. It fires on no frame of the clean held-out cell, on a rounded 0.0 percent of the other clean-visible held-out cells, and on no development night frame under any thermal condition, so on those frames every variant equals its baseline. That matters because the baseline itself is right there. At night, `union` keeps the visible stream that the shipped veto discards. On a night test registered on pohang01's test split, 1,109 frame pairs that no fusion system had scored, `union` beats `shipped` by +0.0273 [+0.0154, +0.0400] and holds the claim at −0.0010 [−0.0049, +0.0031], while `shipped`, which vetoes the visible stream on 90.3 percent of those frames, fails it by 0.0283 (Appendix G.4). The detectors trained on the same run's training split, so this tests the fusion rule, not the detectors, and night AP levels swing from 0.04 to 0.35 between the run's 52-second episodes; directions, not levels, transfer. The gate's firing rate on these 1,109 night pairs has not been measured.

D. An observation, not a test: ranking the adapted streams

IG-BN inherits a weakness of `union`. Where the visible stream is clean, the fused output falls below the visible stream alone, by 0.0085 to 0.0132 on the held-out run, because the thermal stream's boxes are interleaved with the visible stream's. A fixed ordering that ranks every visible box above every thermal box (`lex`) cannot fall below the visible stream: the scorer matches boxes greedily in score order, so a tail of boxes appended below every existing box cannot lower AP (Section VI-C). The same ordering of IG-BN's adapted streams is above IG-BN in all nine held-out cells, by +0.0096 to +0.0134 where the visible stream is clean and +0.0008 to

+0.0067 where it is corrupted, and it holds the claim in every cell. No verdict was registered for it, and on development `lex` averaged 0.0037 below `union` over the 60 cells. The two data sets disagree, so we report it as a candidate for the next registration and recommend IG-BN with `union`, as registered. The visible detector with IG-BN and no thermal boxes at all was not scored as a system.

VI. What did not work on the same system

The adaptation was the last of a sequence of registered attempts to use the second sensor. The earlier ones are what made its design plain, and we summarise them here; the full record is in Appendices D–F.

A. Predicted uncertainty is informative but does not improve the fusion

Each detector carries a Gaussian variance head [20] trained with a beta-weighted negative log-likelihood [35], and the system was designed so that predicted σ would weight the fusion. We pre-registered the comparison that answers whether it should: real σ against the same σ shuffled onto the wrong boxes, at least three of four conditions above 0.0060 with a block-bootstrap interval excluding zero (Appendix E.2). Used as a weight on box coordinates, σ passes on none of four conditions, and the system's fusion weight was measured to take one value on every frame. Used to re-rank scores, real σ beats shuffled σ on three of four conditions, so it carries information about which boxes are right. But that information is within each detector: by day the gain is the visible detector re-ranking its own boxes, and adding the thermal stream moves it by -0.0040 to +0.0035 with no consistent sign. Against the system with no σ at all the re-ranked system is below on six of eight cells. A shuffled control establishes that a signal exists, not that using it helps. Learned jointly with confidence in a re-ranker of the visible boxes, σ adds +0.0050 [+0.0038, +0.0064] on one checkpoint, and its pre-registered replication on the five retrained detectors failed: positive on all five, beyond the floor on two. Two further lessons came from this test. The recorded runs had scored a ship-and-buoy macro instead of the registered ship AP, and the macro read no effect on the re-ranking use where ship AP reads a pass; checking that the scored quantity is the registered one belongs in the registration. And on physically modelled corruptions the re-ranking pass falls to one or two of four conditions.

B. A sensor veto is a claim about the detector

The system that shipped dropped the visible stream when the thermal camera voted night and the visible frame read dark or veiled (Appendix D.2). Its night arm was written while the visible detector scored 0.0000 at night. That zero turned out to be a label artifact: a filter had deleted every box in dark night frames from the training labels, and a registered restore of the boxes took night visible mAP50-95 from 0.0000 to 0.2520 (Appendix D.4). Retrained on the corrected labels, the visible detector sees at night, and the unchanged rule throws it away: on clean night the fused output equals the thermal stream alone and falls 0.1847 [0.1655, 0.2039] below the visible stream (Appendix D.3). Nothing in the image changed; what changed is the claim the rule makes about the detector. The rule gated on darkness, while what decides whether the visible stream should be dropped is its health, and with the retrained detector those variables no longer coincide. The same confusion appears on the thermal side: under a fogged or noisy thermal stream, the veto's fallback reads a dark low-light day as a fogged night and vetoes a working visible stream on up to 96 percent of day frames (Appendix D.5). The adaptation's gate keeps the two questions apart: it asks the thermal camera whether it is day, and each stream whether it is healthy.

C. No gate beat removing the gate

Seven learned or hand-written replacements for the veto were registered and scored on fresh corruption draws, from per-stream calibration and Bayesian pair fusion to gradient-boosted trust over 32 backbone features (Appendix F). None beat switching the veto off: `union` is better than `shipped` on 30 of 60 development cells and worse on none, and every gate was worse than `shipped` somewhere by day. Two mechanisms explain most of it. A model of box correctness fitted to a noisy matching target re-orders a detector's own boxes worse than its confidence does. And a frame-level gate pools frames whose scores live on different scales: switching to the visible stream alone on a random three quarters of frames, a choice that carries no information, scores 0.0085 below the interpolation between the two pure policies. What worked used the scorer's structure. Boxes appended below every existing box cannot lower AP, so demoting a stream is never worse than deleting it, and a fixed tier order holds the floor of the better stream by construction (`lex`, Section V-D). The largest gain needed no fusion idea at all: ensembling frozen checkpoints helps every arm it is put under, and `full_tta` beats `shipped` on 57 of 60 development cells and on all nine held-out cells (Appendix G.3). The adaptation is the one change in this sequence that acts inside a detector, and it adds to the ensemble rather than competing with it.

VII. Discussion and limitations

What the thermal camera contributes, and what is not yet shown. As a second detector scored against the visible labels, the thermal stream adds little by day on this data: on the held-out run it scores 0.0156 against the visible stream's 0.2767, and fusing it lowers the output wherever the visible stream is clean (Section V-D). That comparison is confounded: the held-out run has visible labels only, so a true thermal-only detection counts as a false positive, and against its own labels on development the thermal detector scores 0.141 ship AP by day, against the visible detector's 0.378 on the same frames (Table S6); the 0.05 percent partner rate at IoU 0.85 (Section III-B) may reflect cross-camera box correspondence as much as weak sensing. As a gate, the thermal camera separated a dark scene from a degraded camera on development, where an underexposed visible frame and a night frame look alike to every statistic of the visible frame

we tried. Three things limit how far that goes. First, the held-out run cannot test it: it is all daylight with a clean thermal stream in every adaptation cell. Second, the gains themselves are mostly what BN re-estimation does under a shift [2]; the gate's own value is in the losses it avoids, and that rests on an unregistered screen (Section V-A). Third, a clock could do the day–night part. On this data the night run is entirely night and the day runs entirely day, so a solar-elevation gate from GPS time, or the visible camera's exposure and gain metadata, would separate development day from night as well. The thermal reading would earn its place only where a clock is wrong about the light the camera sees: dusk, dawn, heavy overcast, bridges and covered berths. None of those were tested. The first registration we would add is the ablation that separates these: no gate, the visible term alone, the thermal day term alone, a solar-elevation gate and IG-BN, on development and held-out data.

What we recommend. For a two-stream detector built from frozen detectors: fuse by `union`, without a night veto; gate BN-mix adaptation of the visible detector on the thermal camera's daylight reading and both streams' health (IG-BN), in its causal form (IG-BN-causal) for a live stream; and where ten detector passes per stream are affordable, apply the same adaptation inside an ensemble of frozen checkpoints with flipped views (IG-BN+ens). Every element was registered before it was scored and, under the corruptions it was developed on, replicated on the held-out run. Ranking visible boxes above thermal ones is a candidate, not a recommendation (Section V-D).

Limitations. The full list is in Appendix I; these bound the claim.

- *Simulated corruptions, one stream at a time.* Every gain is on simulated corruptions. The largest, under sensor noise, is on an ImageNet-C-style severity with no physical anchor and is a first test, not a replication. Each held-out cell corrupts one stream with the other clean; correlated degradations such as fog in both bands were tested only on development, where the gate fires less (Section V-A). The gate does not fire on clean held-out frames, so the method has not met a real degradation.
- *One held-out run, all daylight, scored repeatedly.* The held-out run is one daylight run, scored nine times through registered tests, and two gate constants were chosen with its label-free statistics (Section IV-D). Each later result is a repeated look at the same run, so agreement with development shows replication on one run, not generalisation across runs. Its thermal frames sit far from development's (median health novelty 106.4 against 7.9), which the merge bound tolerates; the night behaviour rests on development and on the pohang01 test split, which shares a run with the detectors' training data.
- *Intervals.* Held-out intervals cover frame sampling around the five-system mean, not variance over systems or draws; development intervals are t-intervals over the five systems (Section III-D).
- *Transfer of the thermal reading.* `ir_p05` is read from 8-bit frames stretched per frame, from one camera in one season. A radiometric 14-bit stream, another camera, another season or sea state, or thermal crossover could move it, and the 41.5 threshold would need refitting.
- *A healthy thermal stream is required.* Where the thermal stream is degraded the gate fires less, and every held-out adaptation cell has a clean thermal stream.
- *Development draws were reused.* Five development verdicts share one set of draws, and IG-BN+ens and IG-BN+ens+glare reused the draws of the gate study; each later verdict is weaker evidence than a first test, as its registration states.
- *Cost.* IG-BN reads up to 0.7 s ahead; IG-BN-causal removes that at a second pass per gated block. Neither IG-BN nor IG-BN+ens was timed. The shipped two-stream pipeline runs at 9.1 FPS end to end on a laptop GPU, bound by its NumPy frame statistics (60.6 ms) rather than its two detectors (38.3 ms), and the gate reads the same statistics.
- *Dataset and metric.* Pohang only, a canal and port route with one night run; sun glint, spray and open-sea states are absent. We report AP50-95 only, with no breakdown by target size or range and no recall at a fixed false-alarm rate, which a collision-avoidance system would need.

A note on the protocol. A measured paired noise floor, block-bootstrap intervals and magnitude floors moved 20 of 74 earlier findings of this project to indeterminate when applied retroactively, and a registration exposed a wrong metric in its own recorded runs (Section VI-A). The adaptation's gains survive these corrections by an order of magnitude: the smallest headline gain, +0.0186, is nearly eight times the 0.0024 maximum of the ship-AP noise floor (Appendix C).

VIII. Conclusion

Test-time adaptation of a visible detector is easy to apply and hard to aim: on our development set it helped on degraded frames and hurt on clean and night frames, and a visible camera cannot tell a dark scene from a dark sensor. Gating BatchNorm re-estimation on a co-mounted thermal camera's reading of daylight and on both streams' health kept the gains with no development or held-out cell worse. Pre-registered, it passed development and replicated on one held-out daylight run under fog (+0.037 ship AP), low light (+0.019) and glare (+0.031), and gained +0.18 under sensor noise in a first test; its causal form cost at most 0.0038 on development and 0.0007 on the held-out run, and inside a ten-pass ensemble its gains added under noise and fog and held elsewhere. Whether the thermal camera, rather than a clock, is what the gate needs remains to be tested. The same system's earlier attempts to use the second sensor, as an uncertainty-weighted voter and as a sensor veto, did not survive registration.

Data and code availability. The Pohang Canal images and PoLaRIS labels are public [26, 27]. The code, every pre-registration as a dated commit, the exposure ledger and the result record behind each table and figure (Table S17) are in the public repository (github.com/Laksh-saroha/uqfusion, branch `fusion-uq-phase3`).

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